7.2 Technological Barriers

7.2.1 Power and energy limitations

One of the most persistent and defining challenges in drone evolution has been the limitation of power and energy sources, directly influencing flight duration, payload capacity, and operational efficiency. Despite significant advancements in propulsion systems and lightweight materials, the endurance of most drones remains restricted by battery energy density and power management constraints. Conventional lithium-polymer (Li-Po) and lithium-ion batteries, though efficient and lightweight, still offer limited energy output relative to the power demands of high-performance UAVs. This restricts long-range missions, especially in surveillance, mapping, and delivery applications that require sustained flight over extended periods.

To overcome these limitations, researchers and developers have been exploring alternative energy systems that can enhance endurance while maintaining operational safety. Solar-powered drones have demonstrated promise in achieving prolonged flights by harnessing renewable energy, though efficiency depends heavily on weather and daylight conditions. Similarly, hydrogen fuel cells are emerging as viable solutions, offering higher energy density and cleaner operation compared to conventional batteries. Hybrid propulsion systems combining battery and combustion-based energy sources are also being tested to balance efficiency, power, and sustainability.

Another crucial area of innovation lies in wireless power transmission and energy harvesting technologies, which can potentially allow drones to recharge mid-air or while hovering over charging platforms. Moreover, advances in nanomaterials, solid-state batteries, and intelligent power management algorithms are improving energy storage and utilization efficiency.